CS 412 / 512
Introduction to Computer Graphics

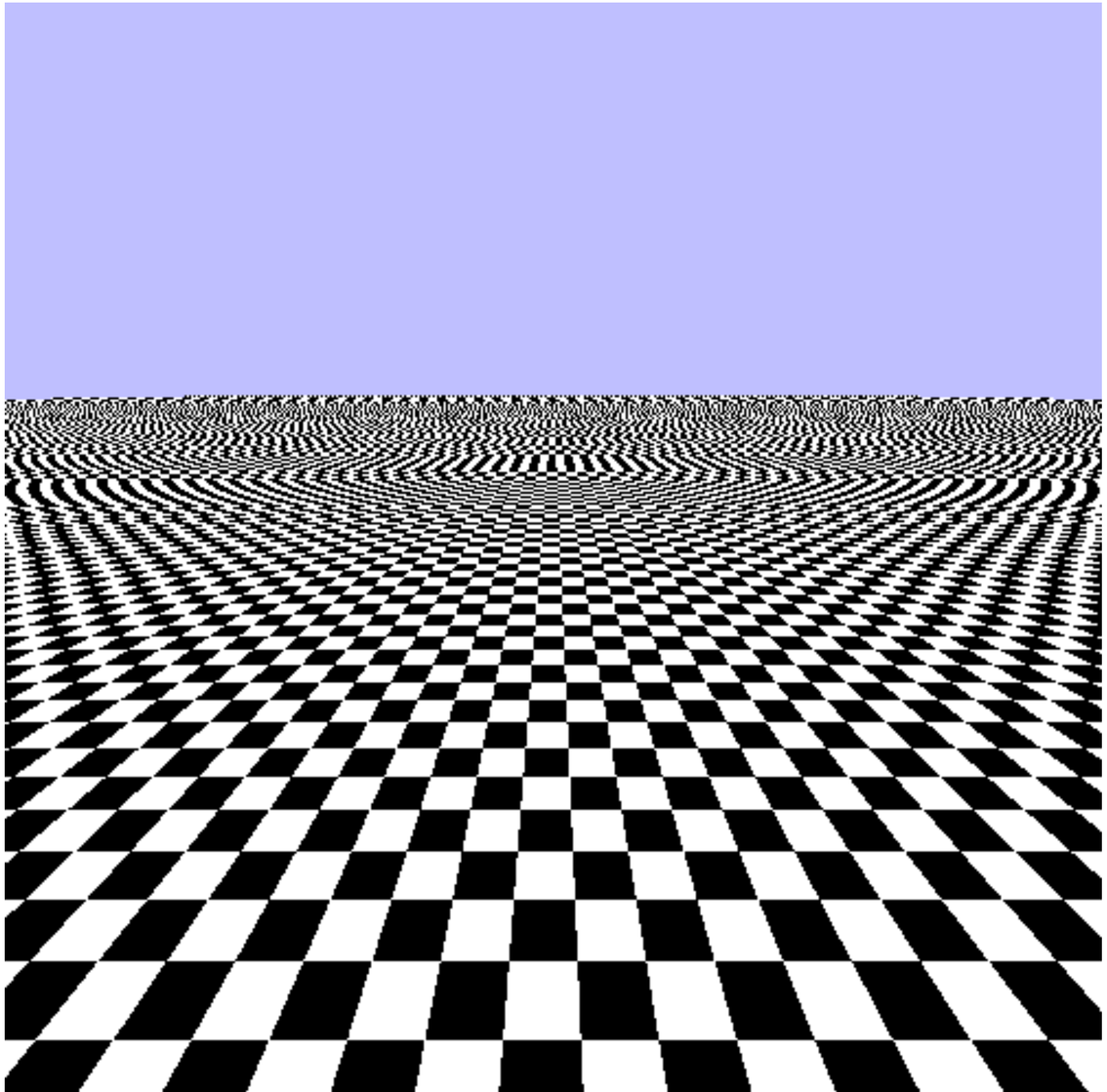
Zhu Wang
Assistant Professor of Computer Science



COMPUTER SCIENCE

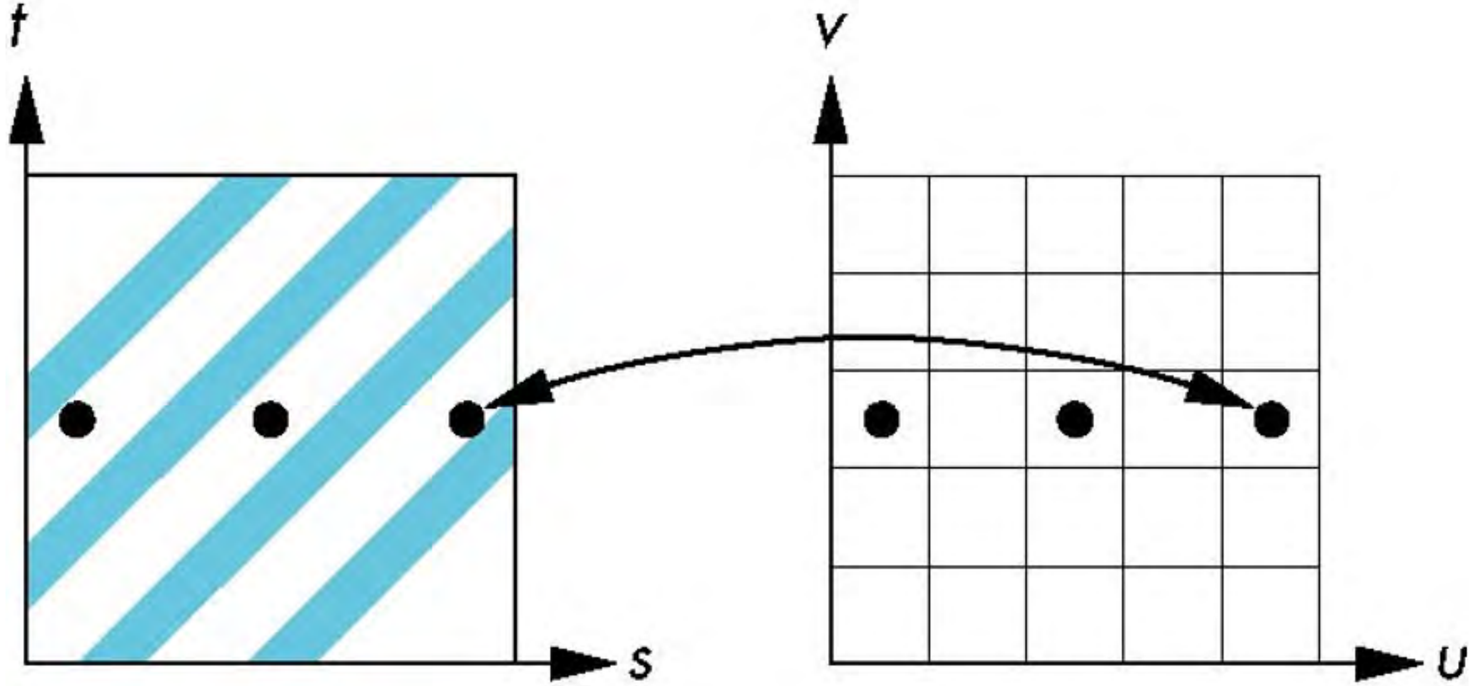
Aliasing, Filtering & Mipmap

Aliasing



[source: Sophismata]

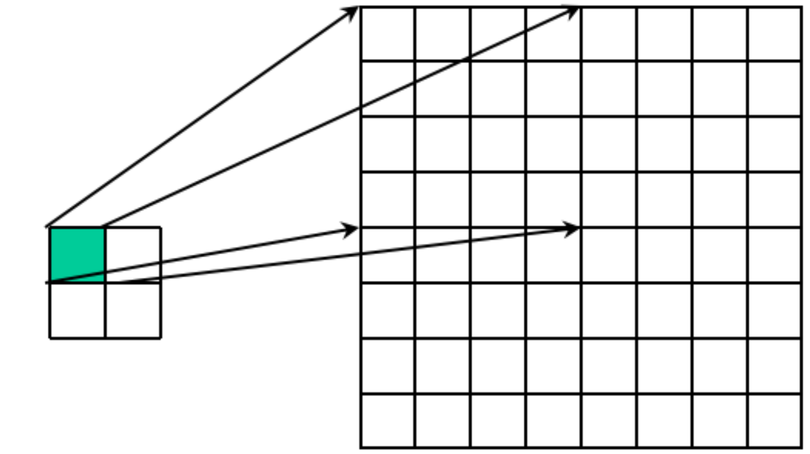
Aliasing



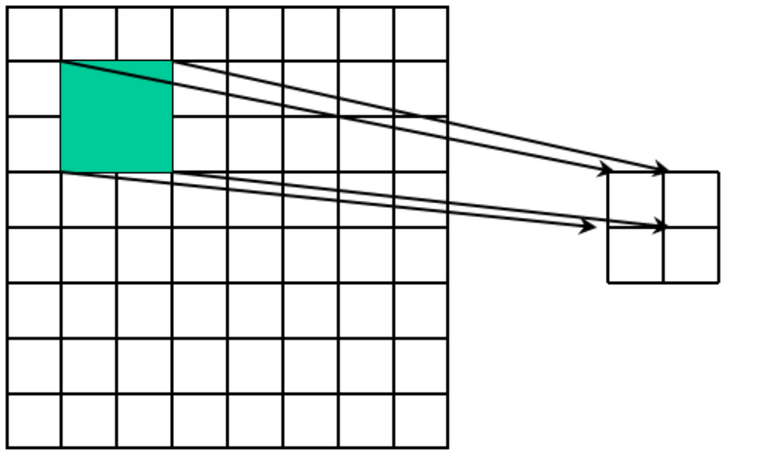
Point sampling of the texture can lead to aliasing errors

[source: Prof. Edward Angel]

Magnification and Minification



Magnification



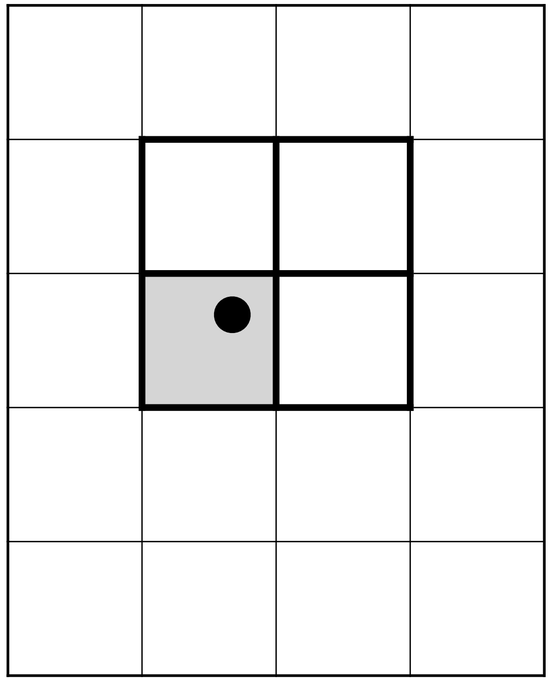
Minification

Magnification: more than one pixel can cover a texel

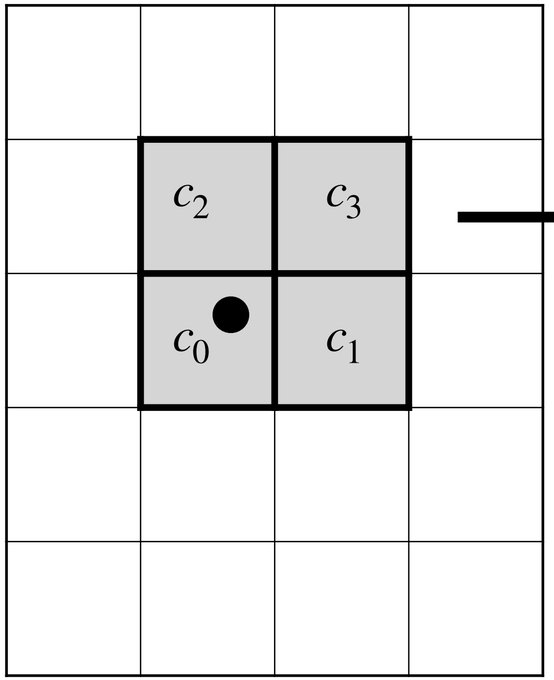
Minification: more than one texel can cover a pixel

[source: Prof. Edward Angel]

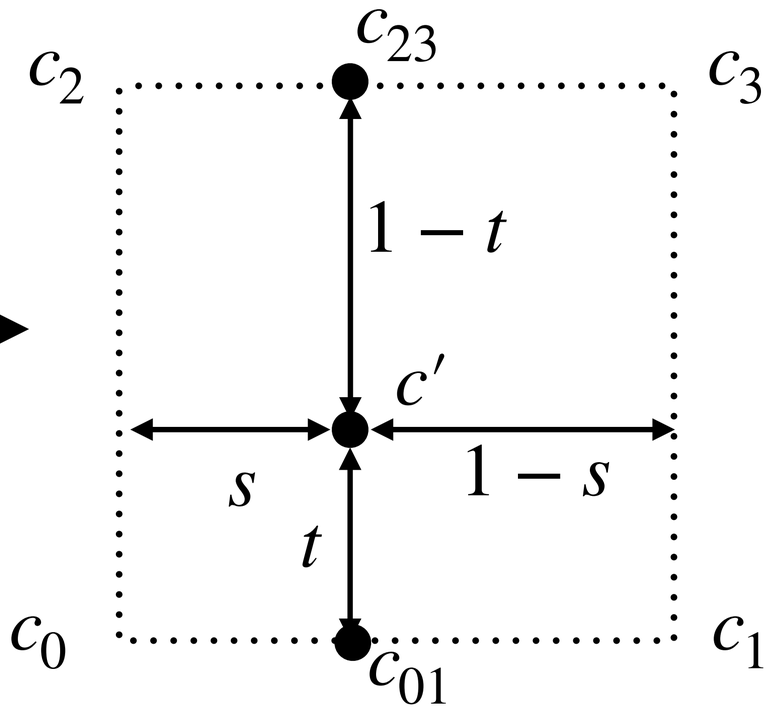
Linear Filtering



Nearest



Linear

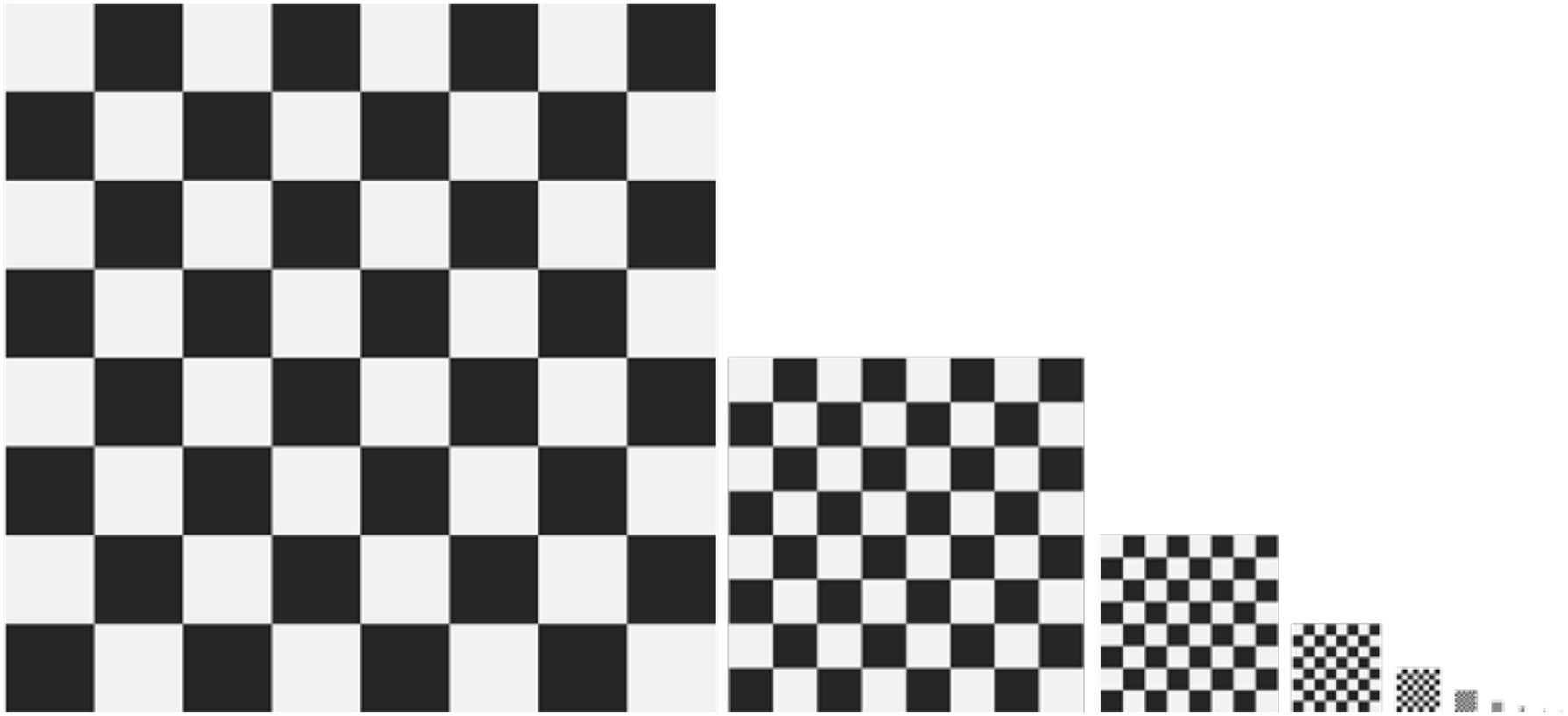


$$c_{01} = (1-s)\cdot c_0 + s\cdot c_1$$

$$c_{23} = (1-s)\cdot c_2 + s\cdot c_3$$

$$c' = (1-t)\cdot c_{01} + t\cdot c_{23}$$

Mipmap

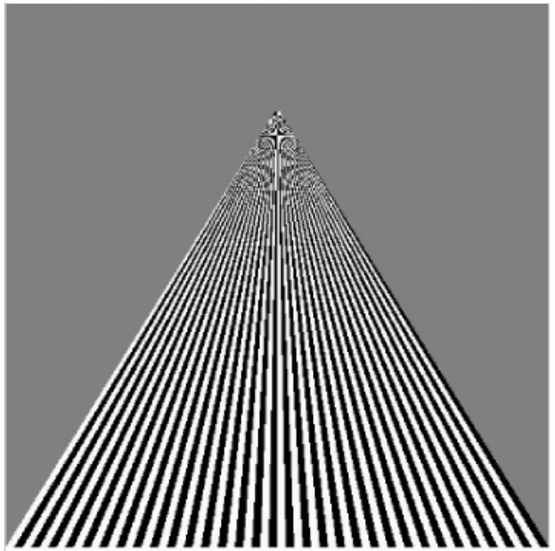


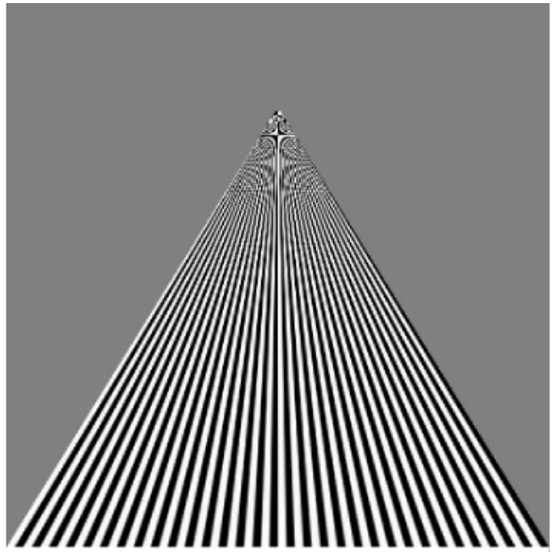
[source: arm developer]

```
gl.generateMipmap(gl.TEXTURE_2D);
gl.texParameteri(gl.TEXTURE_2D, gl.TEXTURE_MIN_FILTER,
                 gl.LINEAR_MIPMAP_LINEAR);
```

Results

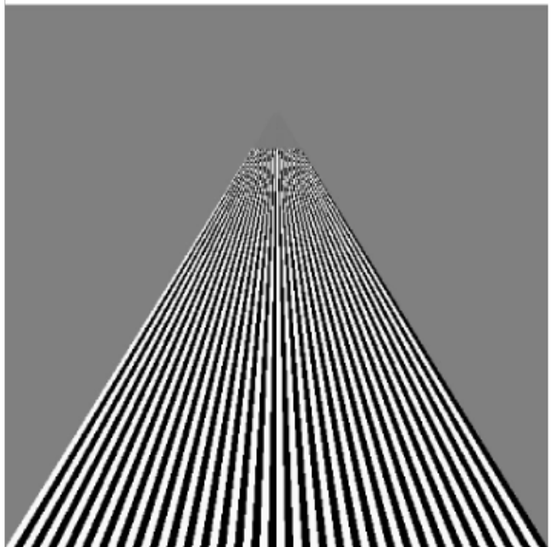
point sampling

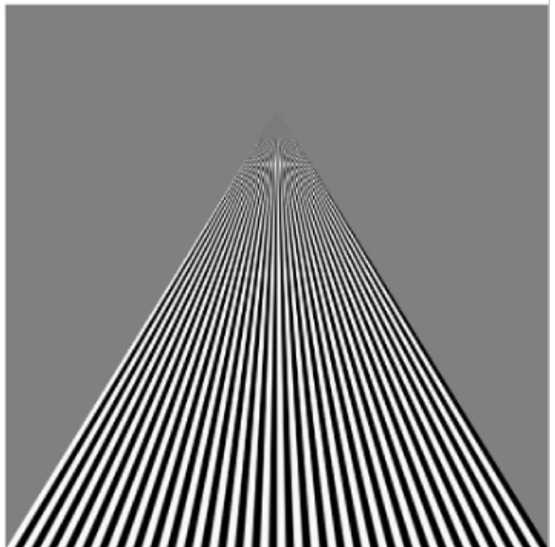




linear filtering

mipmapped point sampling



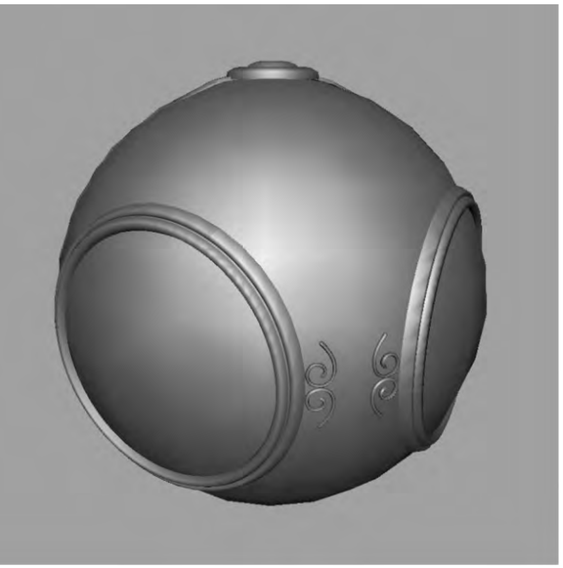


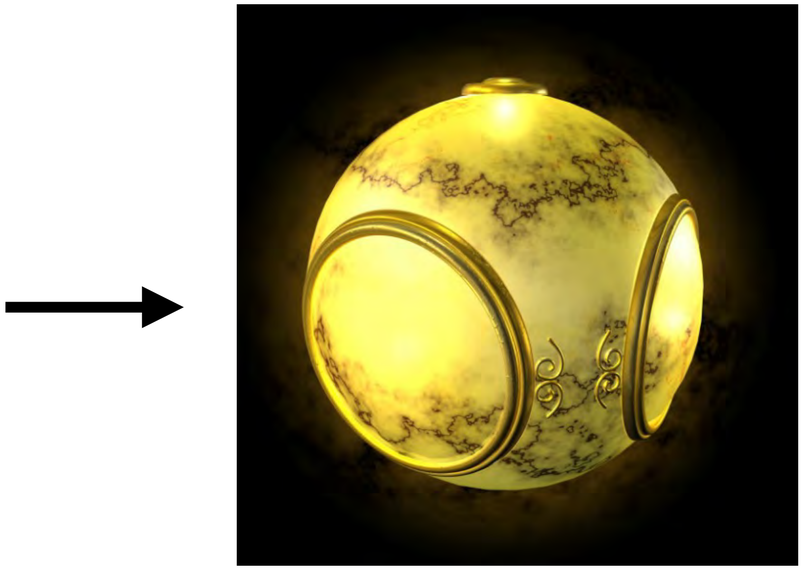
mipmapped linear filtering

[source: Prof. Edward Angel]

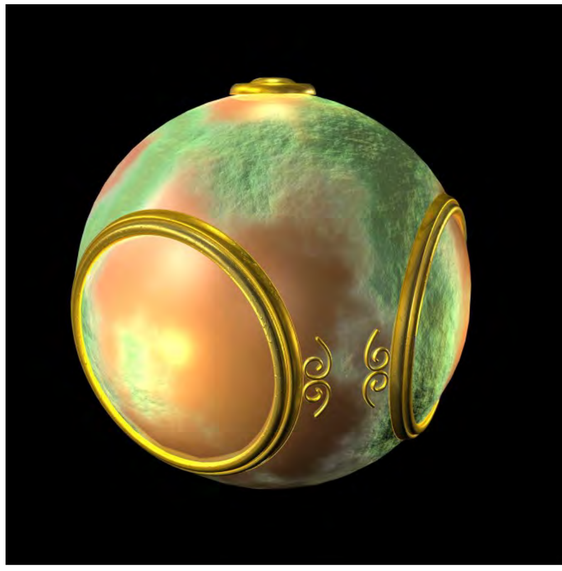
Texture Mapping Techniques

Three Major Types of Texture Mapping









Geometric Model

Texture Mapping

Reflection Mapping

Bump Mapping

[source: Prof. Edward Angel]

Three Major Types of Texture Mapping

Texture Mapping

Uses images to fill inside of polygons

Reflection Mapping (Environment)

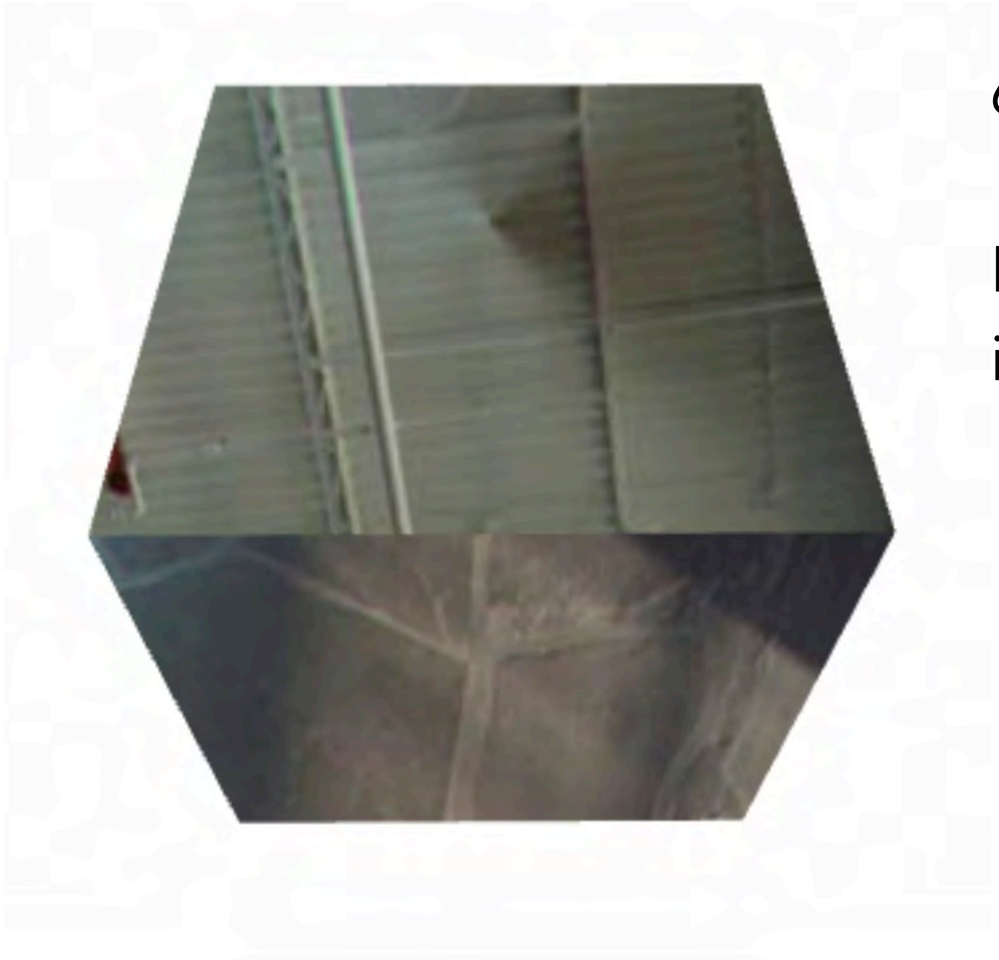
Uses a picture of the environment for texture maps

Allows simulation of highly specular surfaces

Bump Mapping

Emulates altering normal vectors during the rendering process

Environment Mapping



[source: WebGL Fundamentals]

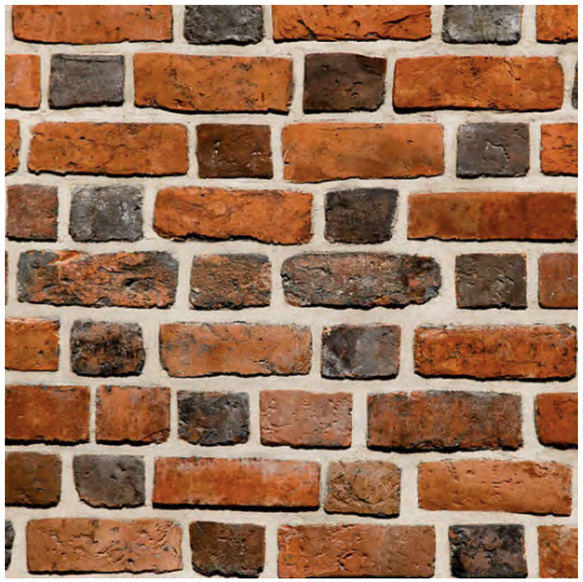
6 textures on 6 faces: +X, -X, +Y, -Y, +Z, -Z

Reflection r determine intersected face and intersection point in 2D coordinates on that face

```
// View direction
vec3 view = normalize(uCameraPos - vWorldPos);
// Surface normal
vec3 n = normalize(vWorldNormal);
// Reflection direction
vec3 r = reflect(-view, n);

// Sample cubemap
vec3 envColor = texture(uEnvMap, r).rgb;
```

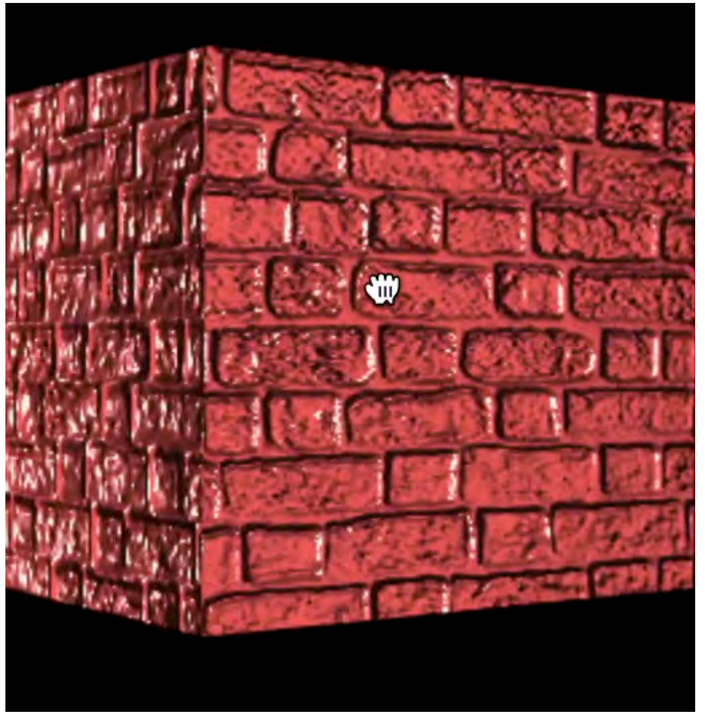
Bump Mapping



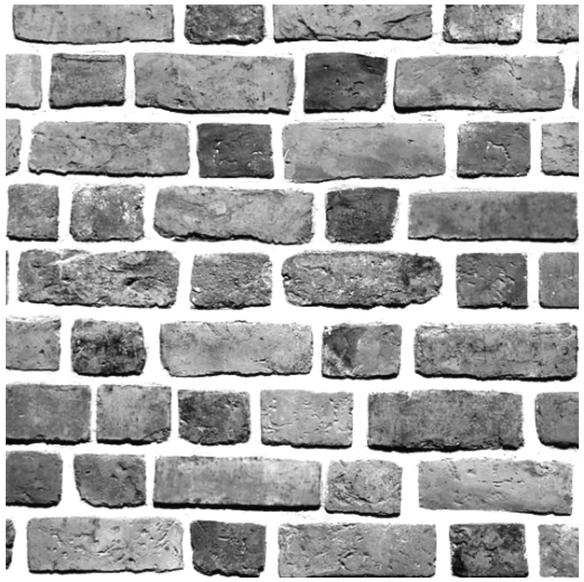
+

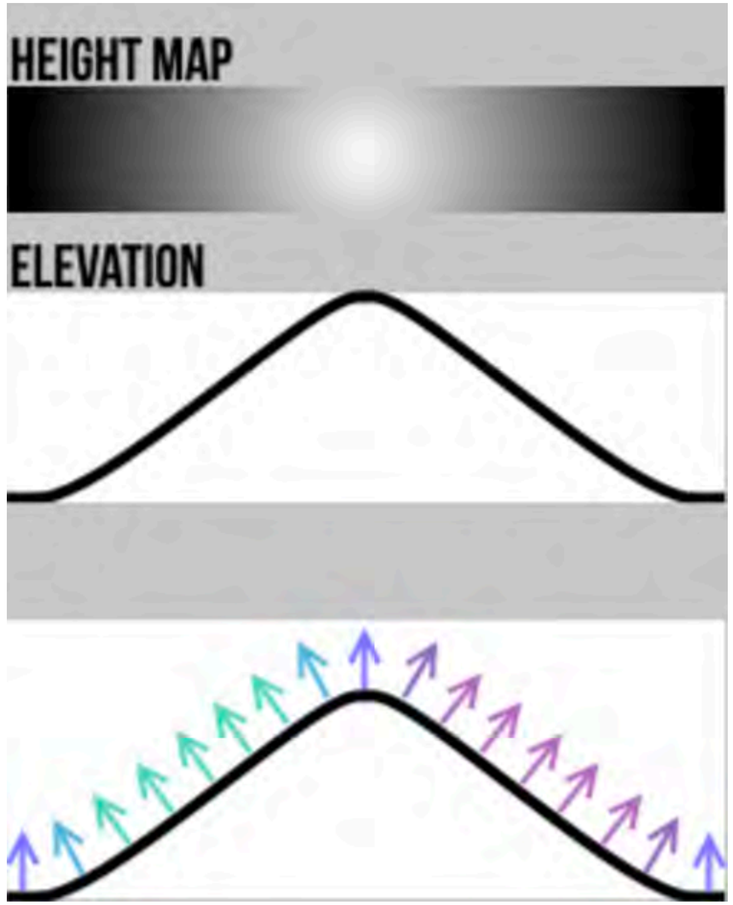


→



Bump Mapping





[source: Blender]

h pixel height

h_u Neighbor in u direction

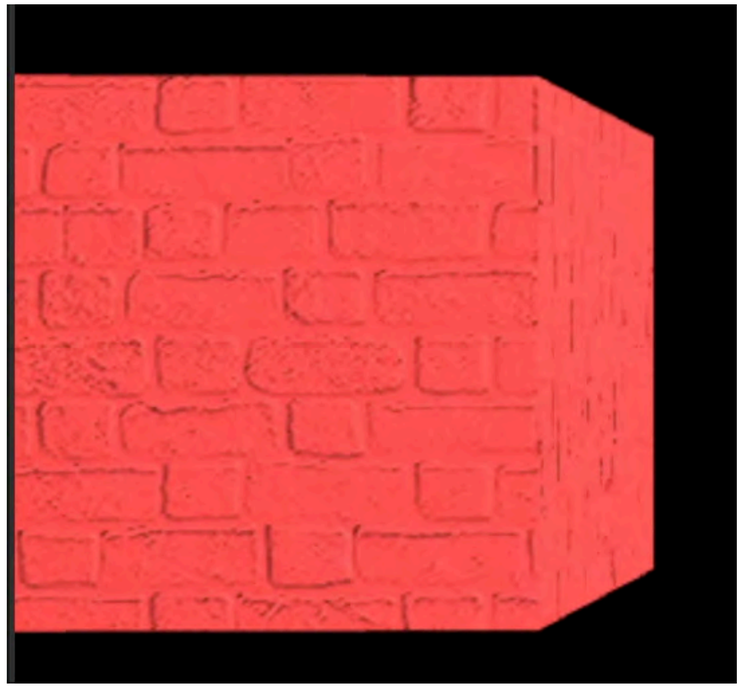
h_v Neighbor in v direction

Gradient

$$\frac{\partial h}{\partial u} \approx h_u - h \qquad \frac{\partial h}{\partial v} \approx h_v - h$$

Bump Mapping

Simple perturbation



```
float h = texture(uTexture, vTexCoord).r;
float hu = texture(uTexture, vTexCoord +
                   vec2(0.01, 0.0)).r;
float hv = texture(uTexture, vTexCoord +
                   vec2(0.0, 0.01)).r;

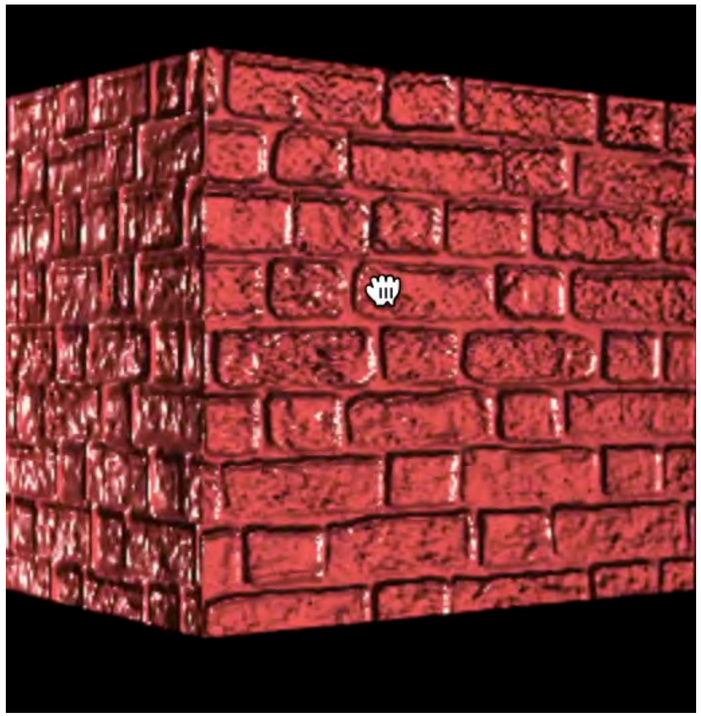
vec3 gradient = vec3(
                   (hu - h) * bumpStrength,
                   (hv - h) * bumpStrength,
                   1.0
                   );
```

Perturb the normals

```
normal = normalize(normal + gradient);
```

Bump Mapping

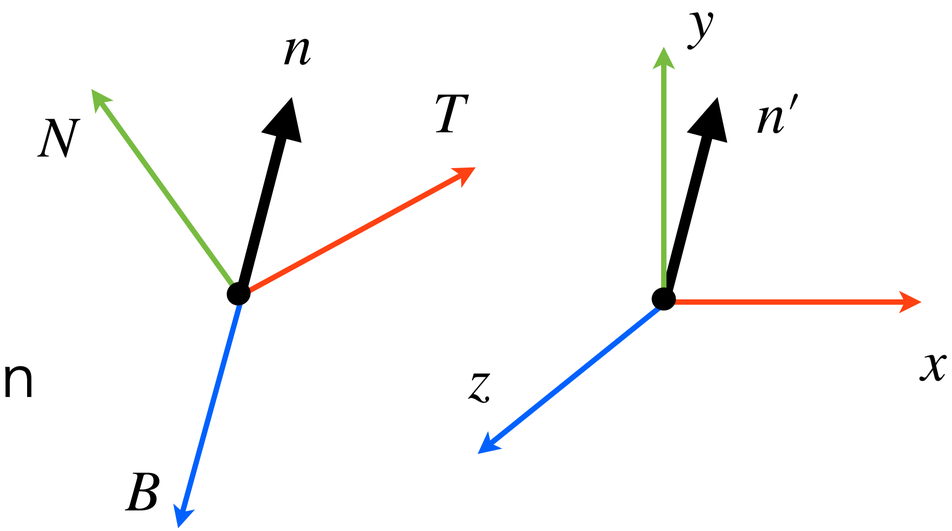
TBN-based normal perturbation



T Tangent: u direction

B Bitangent: v direction

N Surface normal



They form a basis for the tangent space at that vertex

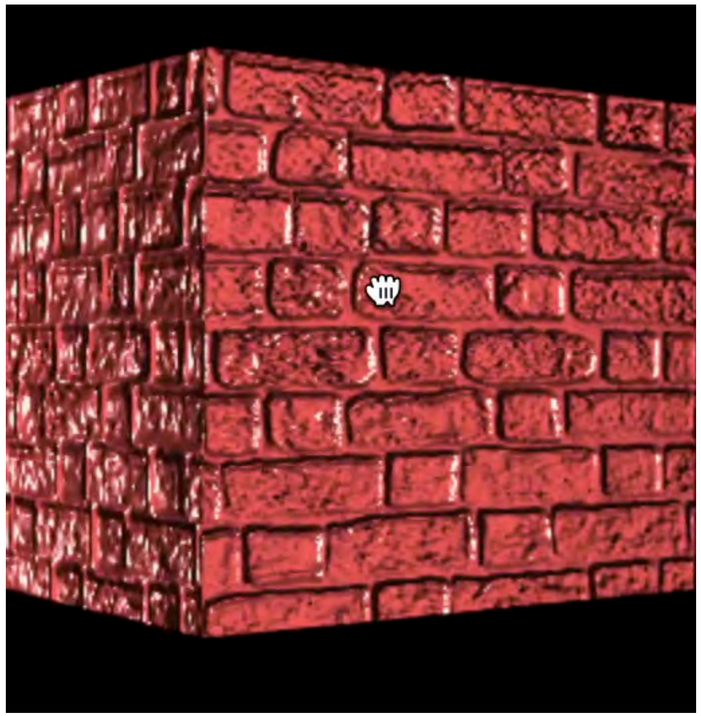
$$TBN = [T\ B\ N] \qquad v_{world} = TBN \cdot v_{tangent}$$

$$v_{world} = xT + yB + zN \qquad v_{tangent} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$n' = TBN \cdot n$$

Bump Mapping

TBN-based normal perturbation



```
float h = texture(uTexture, vTexCoord).r;
float hu = texture(uTexture, vTexCoord +
                   vec2(0.01, 0.0)).r;
float hv = texture(uTexture, vTexCoord +
                   vec2(0.0, 0.01)).r;

vec3 bumpNormal = vec3(
                   (hu - h) * bumpStrength,
                   (hv - h) * bumpStrength,
                   1.0
                   );
```

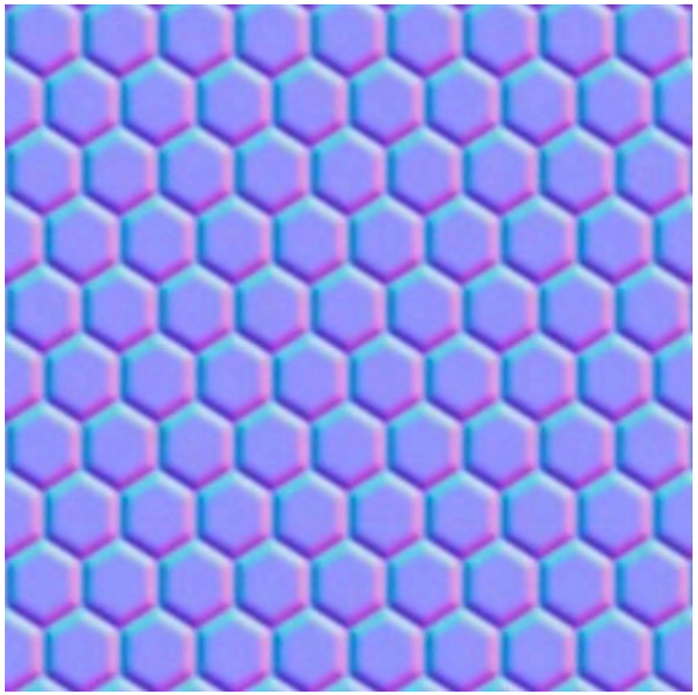
Perturb the normals

```
mat3 TBN = mat3(vTangent, vBitangent, vNormal);
vec3 normal = normalize(TBN * bumpNormal);
```

Heightmap gradient → trangent space normal → world space normal → lighting

Normal Mapping

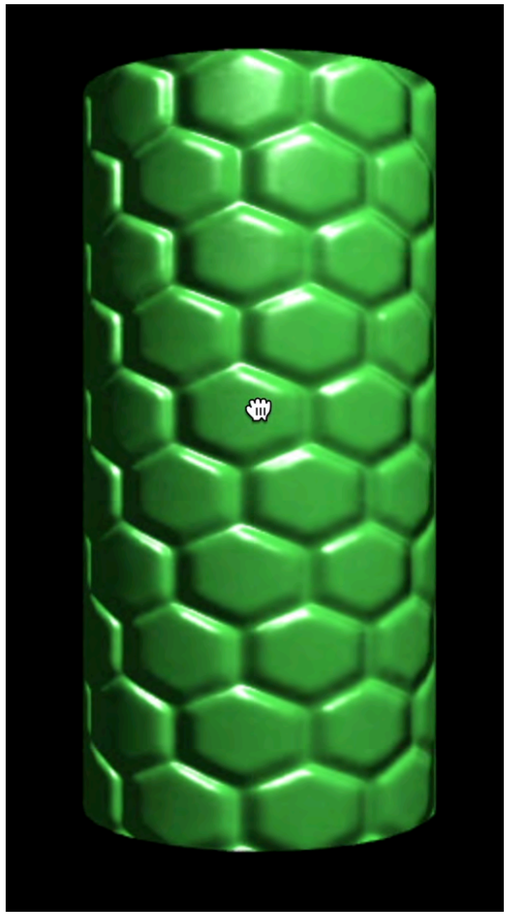
Normal Mapping



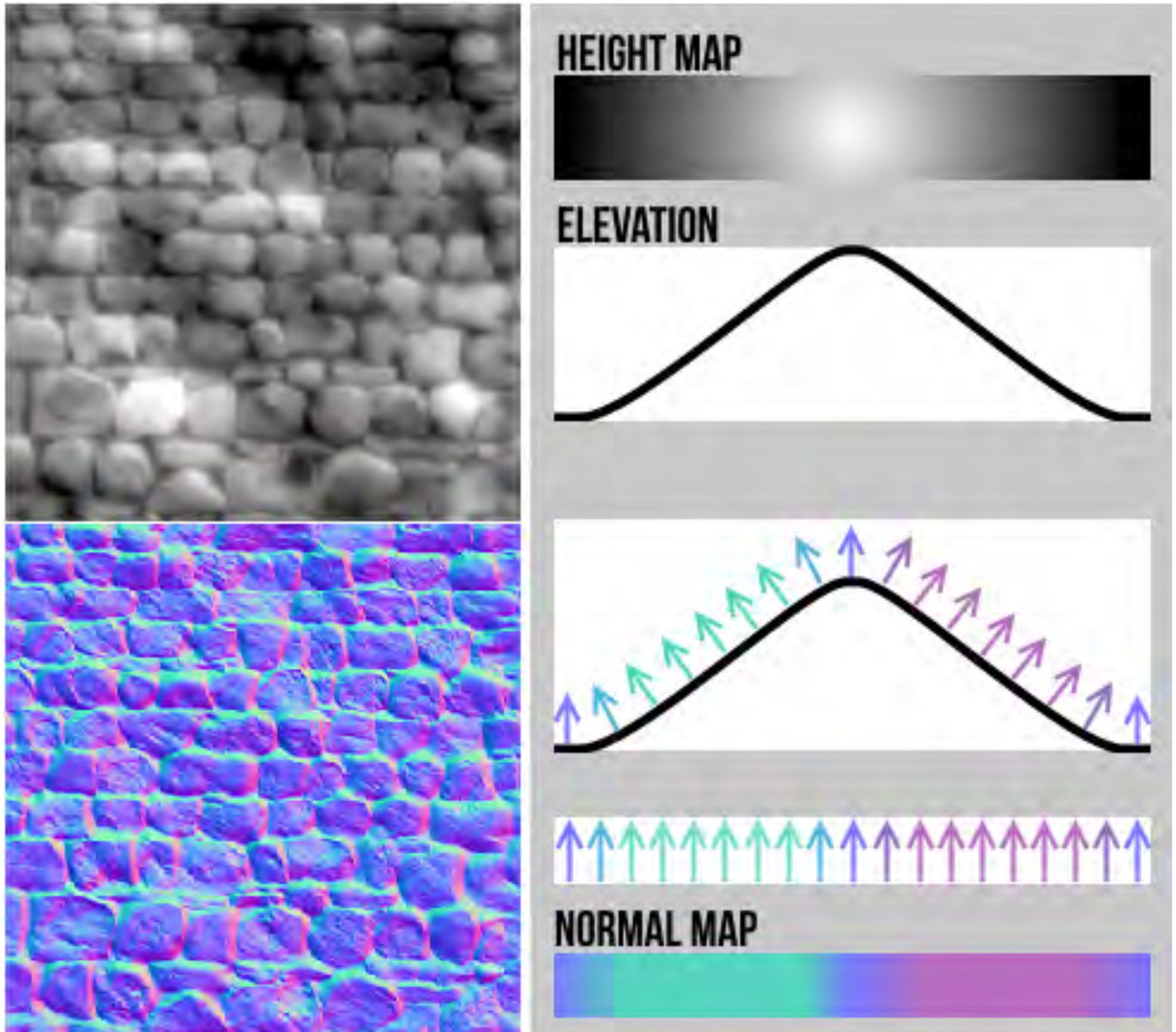
+



→

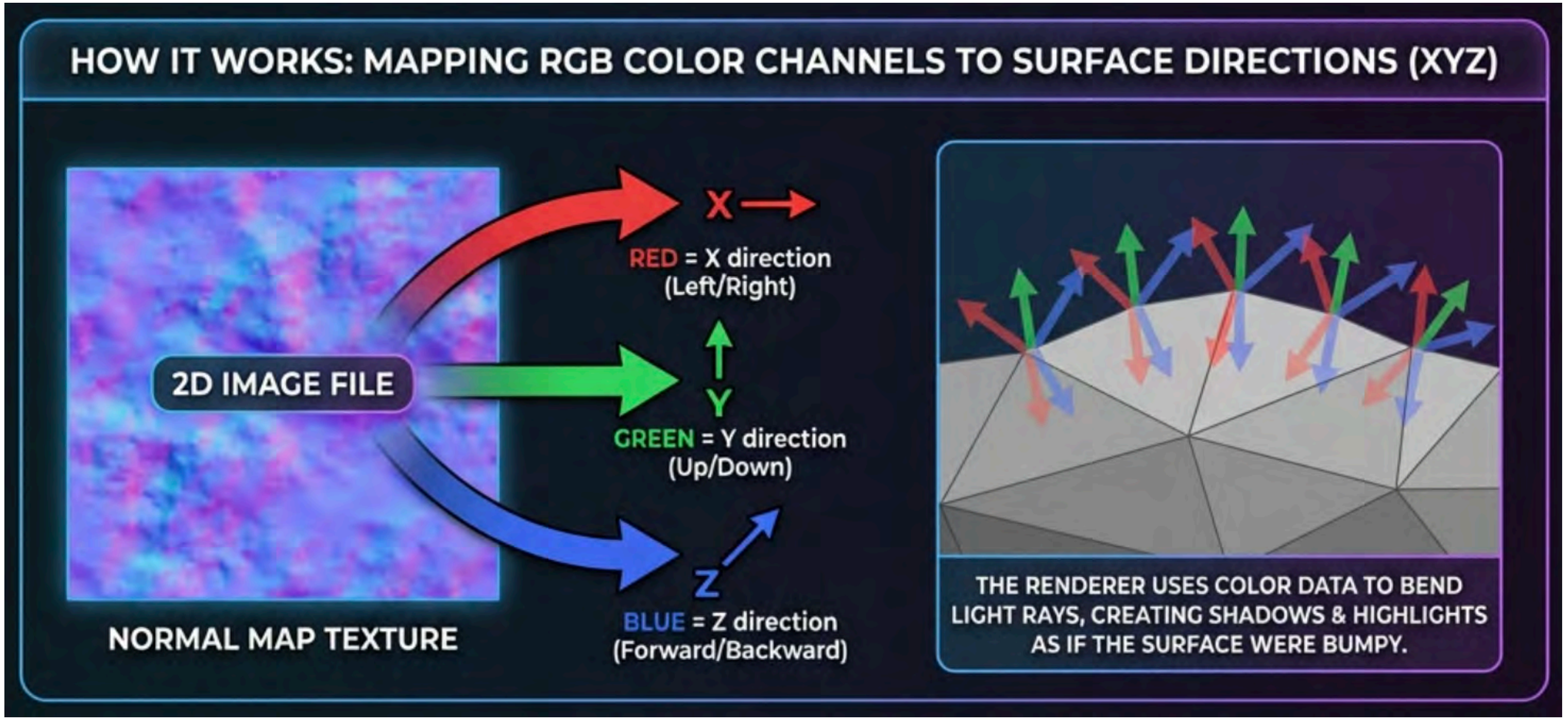


Normal Mapping



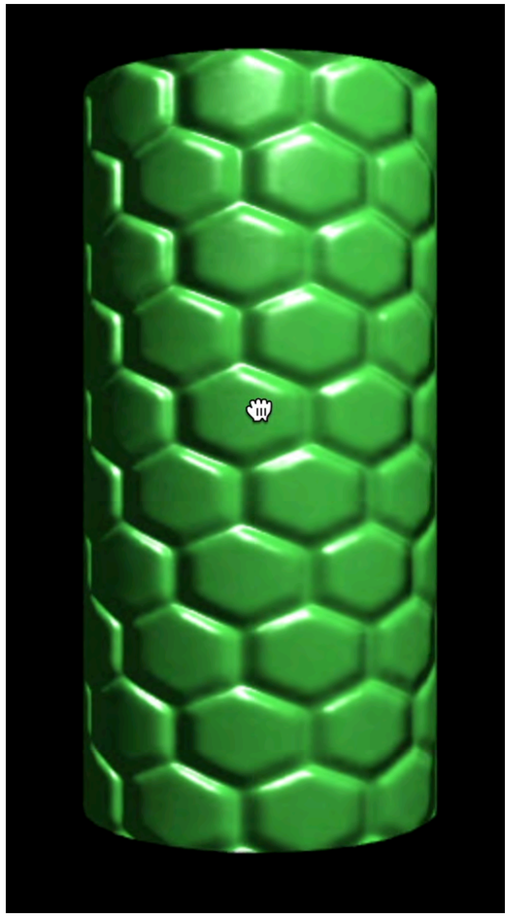
[source: Blender]

Normal Mapping



[source: normalmapgenerator.io]

Normal Mapping



Bump mapping: height in texture → gradient → normal

Normal mapping: normal in texture stored as RGB values

$$n = (R, G, B) \times 2 - 1 \qquad \text{[0, 1] to [-1, 1]}$$

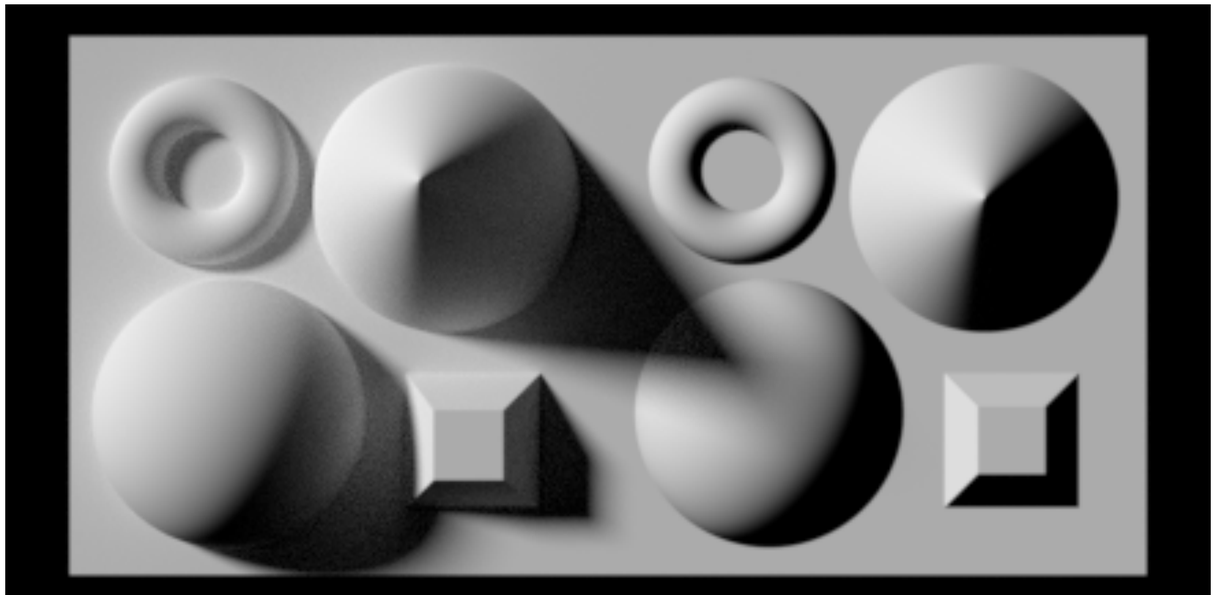
$$TBN = [T\ B\ N] \qquad n' = TBN \cdot n$$

```
vec3 tangentNormal = texture(uNormalMap, vTexCoord).rgb;
tangentNormal = tangentNormal * 2.0 - 1.0;
vec3 worldNormal = normalize(TBN * tangentNormal);
```

Normal Mapping

Normal perturbation doesn’t change actual shape/silhouette





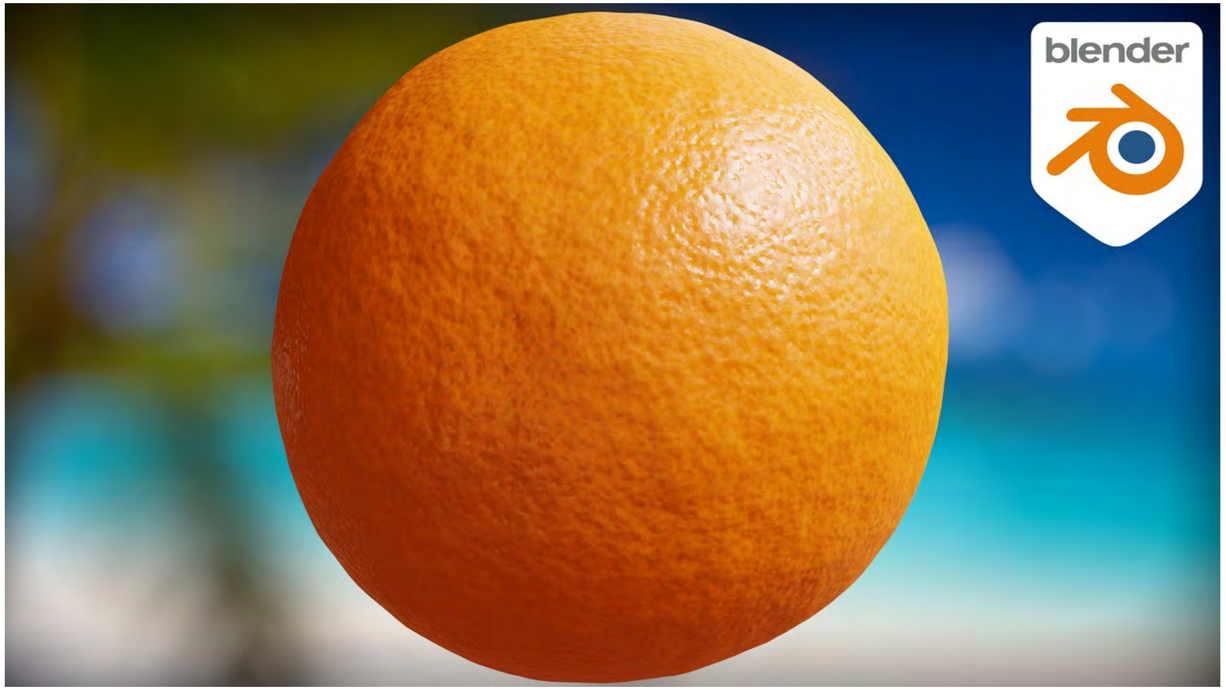
[source: Wikipedia]

Normal Mapping

Better for small details



[source: LearnOpenGL]



[source: Blender]